

AURRA

Spatial Visualization System for Real-Time 3D Interaction

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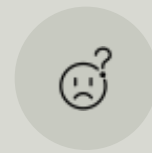


PROBLEM STATEMENT



2D DISPLAY LIMITATION

Current systems rely on flat screens



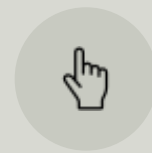
SPATIAL INTERPRETATION

Difficult to interpret 3D data accurately



SLOW DECISIONS

Leads to delayed responses and errors



INTERACTION GAP

Lack of intuitive spatial interaction



AURRA SYSTEM OVERVIEW

SCREENLESS SPATIAL VISUALIZATION

Projection-based display technology

NATURAL INTERACTION

Gesture and stylus-based controls

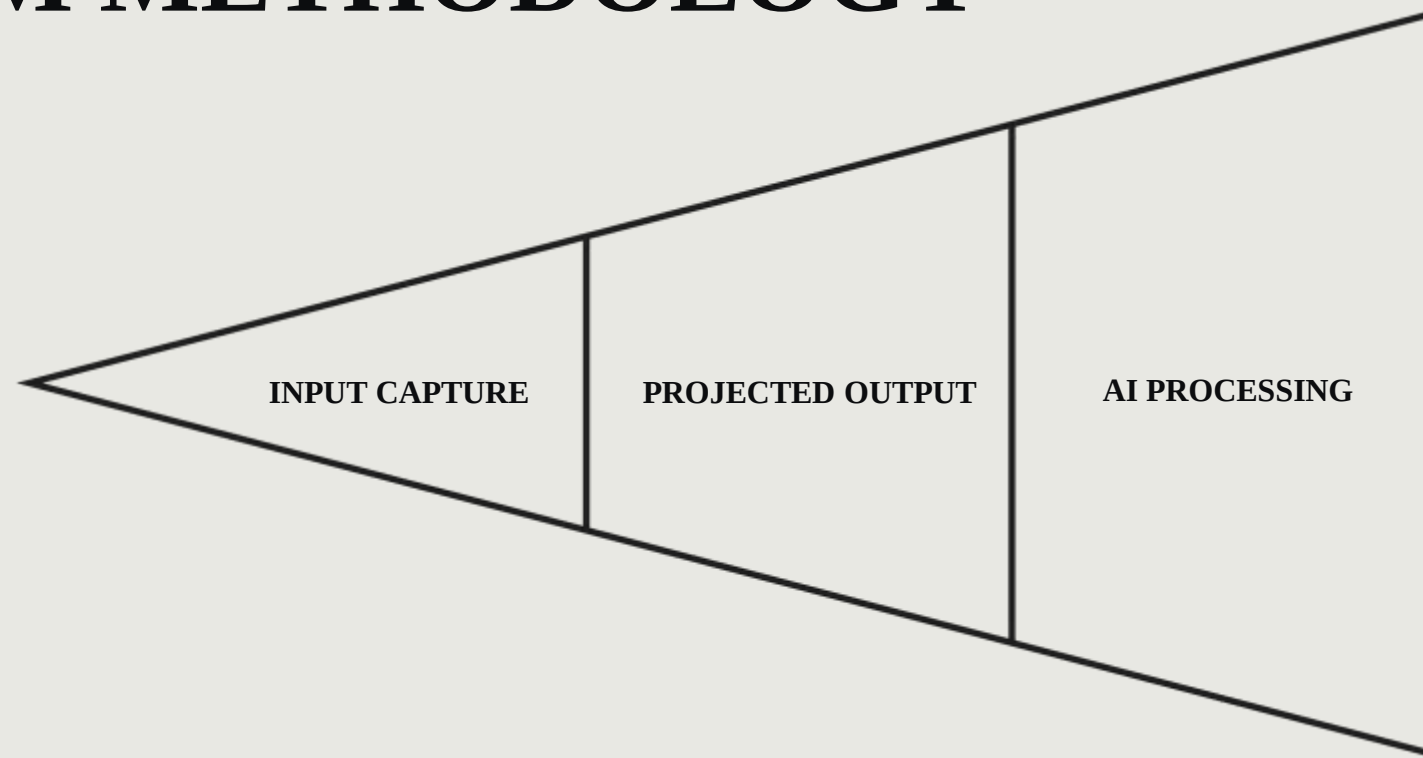
REAL-TIME MANIPULATION

Live 3D model interaction

MULTI-USER COLLABORATION

Simultaneous interaction support

SYSTEM METHODOLOGY



INPUT LAYER

- Depth cameras for gesture tracking
- Multi-sensor array integration
- Real-time user input capture

DISPLAY SYSTEM

- High-lumen projector
- Transparent reflective surface
- Angled spatial projection

PROCESSING CORE

- AI-based interaction system
- Real-time 3D rendering engine
- Multi-user synchronization

PROTOTYPE DESIGN

- Inspired by Pepper's Ghost System and Musion Eyeliner.

EXPECTED RESULTS & IMPACT

1. ENHANCED SPATIAL UNDERSTANDING

Intuitive 3D visualization

2. ACCELERATED DECISION-MAKING

Real-time data interaction

3. ERROR REDUCTION

Improved accuracy through spatial context

4. COLLABORATIVE WORKFLOWS

Multi-user simultaneous operation

5. CROSS-DOMAIN SCALABILITY

Applications in engineering, healthcare, design

REFERENCES & RESEARCH FOUNDATION



3D Holographic Projection – Future of Visual Communication

[IJCSN | Volume 3 | Issue 1](#)



Real-Time 3D Reconstruction using Holographic Telepresence

[MDPI | Applied Sciences | Volume 12 | Issue 8](#)



Holographic Optical Element Research

[Frontiers in Physics | Optics and Photonics | Volume 9](#)



Interactive Holograms using Pepper Ghost Pyramid

[IJSRD | Volume 4 | Issue 1](#)

THANK YOU

AURRA

Augmented Unified Real-Time Rendering Architecture

Any Questions?